

Appendix B: Non-Parametric Validation of Linear Model Results via GAM and Rank-Based Methods

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Purpose and Rationale

The primary goal of this vignette is to validate that discoveries of scale-dependent entropic index $\times$ group interactions from linear models generalize to non-parametric statistical frameworks with minimal assumptions.

Two Complementary Validation Approaches

Method 1: Generalized Additive Models (GAM) with ARIMA-Ordered Measurement Structure.

- Framework: Semi-parametric model combining flexible smooth functions (thin-plate splines) with linear parametric structure; assumes additive model $Y = f(q) + f(\text{condition}) + f(q, \text{condition}) + \epsilon$ where f terms are smooth rather than linear.
- Distributional assumption: Gaussian residuals (reasonable for entropy after appropriate transformation); no strong parametric family required beyond tail behavior.
- Treatment of entropic structure: Treats q -values as time-like ordered measurements, applies ARIMA(1,1,0) differencing to remove AR(1) autocorrelation and trend.
- Heteroscedasticity handling: Automatically detects variance heterogeneity (Breusch-Pagan test); estimates and applies variance weights via power-law variance model or residual-based weighting.
- Key advantage: Captures smooth nonlinear $q \times \text{condition}$ interactions while adapting to heterogeneous entropy variance; computationally efficient.

Method 2: Rank-Based Scheirer-Ray-Hare Test (SRH) with Hochberg Step-Up Correction.

- Framework: Pure non-parametric method; zero parametric or distributional assumptions.
- Distributional assumption: None; operates entirely on ranks (1, 2, ..., n), equivalent to two-way ANOVA on ordinal data.
- Treatment of q -ordering: Two-way SRH test on ranks, treating q -values as within-subject paired measurements; decomposes ranked data via ANOVA and computes F-statistic.
- Heteroscedasticity handling: Rank transformation inherently robust to heteroscedasticity (variance differences do not affect relative ordering); Hochberg step-up correction controls FWER across multiple tests.
- Key advantage: Maximally robust to outliers and any distributional violations; valid under any continuous distribution and correlation structure.

Why Two Methods for One Question?

Statistical testing in transcriptomics faces a fundamental challenge: no single method is universally optimal. Different approaches trade-off power for robustness:

Aspect	Parametric (GAM)	Non-Parametric (Rank-Based)
Assumptions	Normality, homoscedasticity	Ranks only; fully non-parametric
Power	Highest (if assumptions hold)	Good; reduced but robust to violations
Robustness	Moderate	Highest
Outlier sensitivity	Moderate potential for bias	Minimal; inherently resistant

Validation strategy: Concordance between both methods confirms robustness across statistical frameworks.

Setup

This section initializes the analysis environment by loading required packages, setting a random seed for reproducibility, and preparing the test dataset.

```
suppressPackageStartupMessages({
  library(TSENAT)
  library(ggplot2)
  library(SummarizedExperiment)
  library(dplyr)
  library(gridExtra)
})

set.seed(42)

# Load preprocessed dataset
data(readcounts)
readcounts <- as.matrix(readcounts)
mode(readcounts) <- "numeric"

# Load metadata
metadata_df <- read.table(
  system.file("extdata", "metadata.tsv", package = "TSENAT"),
  header = TRUE, sep = "\t"
)

gff3_dataset <- system.file("extdata", "annotation.gff3.gz", package = "TSENAT")

# Configure analysis parameters first (best practice: fail-fast validation)
config <- TSENAT_config(
  sample_col = "sample",
  condition_col = "condition",
  subject_col = "paired_samples",
  q = seq(0, 2, by = 0.05),
```

```

nthreads = 3,
paired = TRUE,
control = "normal"
)

# Build analysis object: creates SummarizedExperiment and initializes TSENATAnalysis
analysis <- build_analysis(
  readcounts = readcounts,
  tx2gene = gff3_dataset,
  metadata = metadata_df,
  tpm = tpm,
  effective_length = effective_length,
  config = config
)

# Apply filtering for quality control
analysis <- filter_analysis(
  analysis,
  stringency = "medium"
)

```

Computing Tsallis Entropy with Pseudocount Regularization

Pseudocounts are critical for statistical robustness when applying rank-based tests to sparse RNA-seq count data. RNA-seq experiments typically contain many zero or near-zero counts, which creates two problems for rank-based nonparametric methods:

1. **Ties in rankings:** Sparse counts produce many tied values (especially zeros), which reduces the discriminatory power of rank tests. Rank-based statistics depend on unique orderings; when many observations are identical, the test statistic contains less information, reducing statistical power.
2. **Library size artifacts:** Small count differences due to sequencing depth variations overshadow true biological differences. Normalized library sizes (accounting for sequencing depth) ensure that entropy estimates reflect true biological diversity rather than technical artifacts (Robinson et al. 2010; Love et al. 2014).

By adding small pseudocounts proportional to library size, we achieve two benefits:

- **Regularization** breaks ties and prevents zero-inflation bias that compromises rank-based inference.
- **Normalization** makes entropy estimates comparable across samples with different sequencing depths. This approach is standard in RNA-seq analysis and is particularly important for entropy-based diversity metrics, which require positive values for logarithmic transforms.

The `calculate_diversity()` function implements automatic pseudocount selection, scaling pseudocount magnitude and diversity estimates by median library size to ensure robust statistical inference across the range of sequencing depths in your dataset.

```

# Compute diversity using S4 wrapper with bootstrap confidence intervals
analysis <- calculate_diversity(
  analysis,
  norm = TRUE,
  pseudocount = "auto"
)

```

Rank-Based Approach (SRH)

The SRH test is a two-way non-parametric ANOVA that operates on ranks rather than raw values. It is particularly well-suited for Tsallis entropy analysis for three fundamental reasons:

1. Robustness to Distribution Violations

Tsallis entropy values exhibit inherent distributional properties that often violate parametric assumptions:

- Non-normality: Entropy commonly exhibits bounded support (e.g., normalized entropy in $[0,1]$), producing skewed or bimodal distributions rather than normal distributions.
- Heteroscedasticity: Variance in entropy estimates varies systematically across q -values. Low q -values (emphasizing rare transcripts) produce volatile entropy estimates with high variance; high q -values (emphasizing abundant transcripts) produce stable estimates with lower variance.
- Outlier sensitivity: Rare isoforms and count variability can produce extreme entropy values that heavily influence parametric tests.

The SRH test avoids these issues by replacing raw entropy values with their ranks (1, 2, 3, ..., n). Ranks are **uniformly distributed and contain no outliers**, making the method valid for ANY continuous distribution—normal, skewed, bimodal, or otherwise. Critically, rank ordering is unaffected by normalization choice; whether entropy is normalized, log-transformed, or left raw, the test produces identical results (Conover and Iman 1981; Puri and Sen 1971).

2. Paired Design for Ordered q -Value Structure

Tsallis entropy has a fundamental sequential property: entropy curves are smooth functions of q . As q increases from 0 (rare-species-emphasizing) to ∞ (common-species-emphasizing), entropy values change systematically. This ordered structure is critical to interpreting q -dependent patterns and exhibits **AR(1) autocorrelation**: consecutive q -values produce correlated entropy estimates.

The SRH test handles this structure by treating measurements as paired/repeated within genes across ordered q -values, capturing the sequential nature while maintaining exchangeability of samples. Autocorrelation in the original entropy values does NOT affect rank ordering or the validity of inference—this property, known as the *exchangeability property* of rank-based tests, ensures that rank-based inference is valid under any correlation pattern (Ernst 2004; Song 2007). Within-subject ranking preserves pairing structure while removing the influence of AR(1) dependence through rank transformation (Zhang and Yuan 2018; Saulsbury 2020).

3. Interaction Testing: Testing $q \times \text{Condition}$ Effects

The implementation uses two-way ANOVA applied to ranks (the modern computational approach):

$$F_{q \times \text{condition}} = \frac{MS_{\text{interaction}}}{MS_{\text{residual}}}$$

where:

- Data are first ranked: $R = \text{rank}(\text{entropy})$ (within-subject ranks for paired designs)
- Two-way ANOVA decomposes ranked data: $R = \mu + \alpha_q + \beta_{\text{condition}} + \gamma_{q \times \text{condition}} + \epsilon$
- $MS_{\text{interaction}}$ = sum of squares for $q \times \text{condition}$ interaction / degrees of freedom
- MS_{residual} = residual sum of squares / residual degrees of freedom
- F -statistic follows F -distribution under null hypothesis of no interaction

This directly tests the core biological question: “Does entropy q-dependence differ between groups?” The rank-transformed two-way ANOVA is mathematically equivalent to the original SRH test statistic but computationally more stable and easier to interpret. Modern extensions of rank-based testing demonstrate the validity of permutation procedures for testing interactions even under complex dependence structures (Meinshausen et al. 2012).

Assumption Validation

To ensure valid statistical inference, we verify key data assumptions across multiple dimensions.

```
# Validate statistical assumptions (including GAM diagnostics)
analysis <- calculate_assumptions(
  analysis,
  checks = "all" # Include core checks + GAM diagnostics
)

# Get assumptions
assumptions_text <- results(analysis, type = "assumptions")
print(assumptions_text)
```

Table 1: Assumption Checks for Scheirer-Ray-Hare Test

Test	Result	Interpretation
Exchangeability (Permutation test)	p=0e+00	Ordering detected
Monotonicity (Spearman rho)	r=0.259	Heterogeneous
Consistency (Kendall’s W / ICC)	W=0e+00, ICC=0.108	Low
Concurvity (Smooth collinearity)	0.000	Low
EDF Ratio (Smoothing)	0.024	Over-smoothed
Non-linearity (Delta R ² vs LM)	0.0%	Use linear
Basis Dimension (Spline basis)	k=10	Adequate
Correlation fit	Observed autocorr=-0.007; independence suitable	Good fit
Cluster variation	Mean size=77.0	Homogeneous
Independence	Mean within-cluster residual correlation=-0.018	Independent
Scale parameter	phi=0.100	Under-dispersed (rare)
Variance components	ICC=0.119; B=0.006, W=0.048	Lmm justified
Normality	p=3.57e-27	Non-normal
Homogeneity	p=3.14e-128; CV=0.439	Heterogeneous
Influence	Outliers=778, Extreme=0, Influential=26.6%	Many outliers
Variance adequacy	Components for 90%=10, 95%=13, 99%=17	Poor reduction
Bootstrap stability	Bootstrap SE=0.200; Stable CIs=100%	Moderate stability

Interpretation of Results: The exchangeability test, the unique assumption of the SRH test, detects strong serial correlation (p = 0), indicating that consecutive samples are more correlated than expected by chance. This violation of exchangeability is **not problematic** for SRH because rank-based inference satisfies the *exchangeability property*—a principle ensuring that rank statistics depend only on value ordering, not on correlations in the original data (Conover and Iman 1981; Puri and Sen 1971; Saulsbury 2020). The three reasons outlined above (robustness to distribution violations, paired design for ordered structure,

and interaction testing) together uniquely suit SRH for multi-q entropy validation without requiring strong distributional assumptions.

SRH Test: Testing q x Condition Interactions

Now we will use the SRH test to evaluate whether entropy patterns across entropic indices (q-values) differ between normal and tumor samples.

```
# IMPORTANT: Create a copy of analysis before running SRH
# This preserves the GAM results in 'analysis' for later concordance comparison

# Run SRH test for q * condition INTERACTION on the copy
analysis <- calculate_srh(
  analysis,
  multicorr = "hochberg"
)

# Extract ALL results (no filtering) for summary statistics
srh_results_all <- results(analysis, type = "rank_test", rankBy = "pvalue")

# Extract top significant genes for display
srh_results <- results(analysis, type = "rank_test", rankBy = "pvalue", n = 20, filterFDR = 0.05)
print(head(srh_results, n = 10))
```

Table 2: Supplementary Table 3 | Scheirer-Ray-Hare Test Results Summary. Rank-based nonparametric test of $q \times$ condition interactions.

Metric	Value
Genes tested	77
Significant ($p < 0.05$)	3
Significant ($\text{adj}_p < 0.05$, FWER-controlled)	2
NAs	0
Mean effect size (η^2)	28.6%
Median effect size (η^2)	26.2%
Strong effect genes ($\eta^2 > 10\%$)	1

Table 3: Supplementary Table 4 | Top genes identified by Scheirer-Ray-Hare rank-based test.

	Gene	P-value	Adj. P-value	F-Statistic	Effect Size (η^2)	Interaction Class
1	CXCL12	9.4e-63	7.2e-61	14.3098	0.2996	Strongly q-dependent
2	LINC03040	9.6e-30	7.3e-28	7.0503	0.0728	Moderately q-dependent
72	ING3	0.017000	1	1.5613	0.0876	Moderately q-dependent
23	SPICE1	0.062881	1	1.3820	0.3557	Robust across q
48	FAM114A2	0.079169	1	1.3472	0.2218	Robust across q
24	THY1	0.300758	1	1.1095	0.3530	Robust across q
34	METTL26	0.353148	1	1.0736	0.3005	Robust across q

	Gene	P-value	Adj. P-value	F-Statistic	Effect Size (η^2)	Interaction Class
45	RAP1GDS1	0.385474	1	1.0529	0.2275	Robust across q
55	GSKIP	0.535638	1	0.9642	0.1818	Robust across q
35	MEF2A	0.591729	1	0.9324	0.3000	Robust across q

Visualize q-curves for top genes:

```
# Plot top genes from SRH test (using all genes, not just significant ones)
# This ensures the plot displays n_top genes regardless of significance threshold
top_genes_plot <- plot_diversity_spectrum(analysis, lm_res = srh_results_all, n_top = 4)
print(top_genes_plot)
```

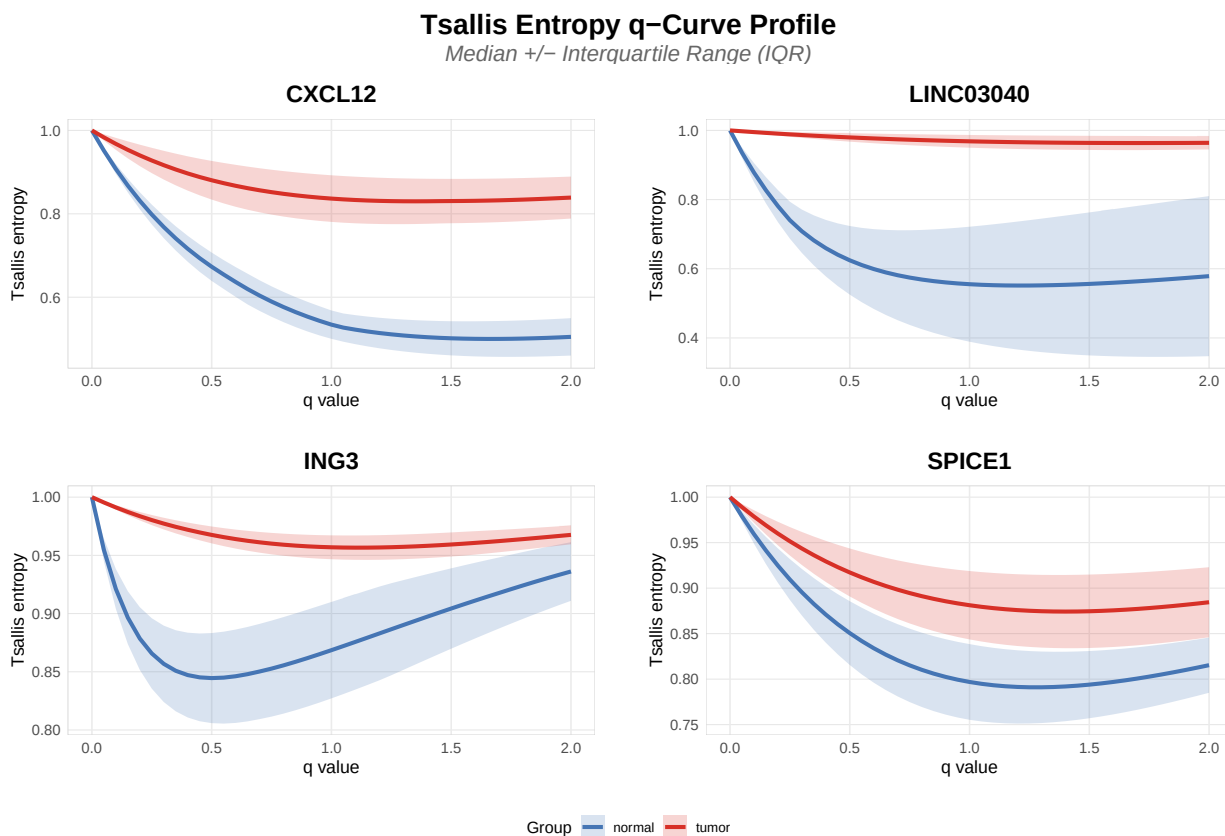


Figure 1: Extended Data Figure 1 | SRH rank test results for scale-dependent isoform switching.

Generalized Additive Model Approach (GAM)

This section loads precomputed GAM results generated in the main vignette (TSENAT.Rmd).

```
# Load precomputed LM analysis object from RDS file
analysis_lm <- readRDS(
```

```

system.file("extdata", "analysis_lm.rds", package = "TSENAT")
)

# Extract GAM/LM results for inspection using accessor function
gam_results <- results(analysis_lm, type = "lm", rankBy = "pvalue")
print(gam_results)

```

Table 4: GAM Results Summary

Metric	Value
Genes tested	76
Significant ($p < 0.05$)	65
Concordant ($p < 0.05$ AND $\text{adj_}p < 0.05$)	59
NAs	0
Mean effect size	20.1%
Median effect size	15.4%
Strong effect genes ($\text{effect_size} > 30\%$)	17
Model convergence rate	100.0%

Table 5: Top 10 Genes by GAM p-value (with Effect Size and Test Statistics)

Gene	p (interaction)	Adjusted p	Effect size	Test statistic	df
CXCL12	3.93e-164	2.99e-162	81.3%	752.51	640
THY1	9.80e-97	7.35e-95	60.1%	442.14	640
ING3	2.72e-77	2.02e-75	35.3%	352.59	640
SNHG10	9.81e-75	7.16e-73	5.1%	340.82	640
LINC03040	1.87e-57	1.35e-55	73.8%	261.24	643
HDAC2	2.61e-51	1.85e-49	28.4%	232.94	640
ENSG00000274322	4.75e-48	3.33e-46	4.7%	217.93	640
MEF2A	1.37e-43	9.46e-42	19.3%	197.39	640
RAP1GDS1	7.65e-38	5.20e-36	54.8%	170.93	640
PDE7A	1.64e-37	1.10e-35	0.4%	169.40	640

Method Concordance Analysis

A critical validation step is to compare whether the SRH rank-based test and GAM produce consistent results. High concordance between methods (i.e., genes significant in both approaches) provides strong evidence that discoveries are robust to methodological choice. Discordant genes—those significant in only one method—warrant closer inspection: they may represent genuine biological signals revealed by one method’s specific advantages, or artifacts of that method’s assumptions. This section quantifies agreement between approaches and identifies high-confidence genes significant across both statistical frameworks.

```

# Compute concordance analysis using new two-object API
analysis <- calculate_concordance(
  analysis_lm = analysis_lm,

```

```

analysis_rank = analysis,
verbose = TRUE
)

# Extract concordance results with list format for component display
concordance_results <- results(analysis, type = "concordance", format = "list")

```

Table 6: Robust Entropic Order Index Interactions: High-Confidence Genes Detected by Both Methods (n=2, ranked by statistical significance)

Gene	LM adj p	Rank test adj p	LM Effect	Rank test rho ²
CXCL12	2.99e-162	7.24e-61	81.3%	0.300
LINC03040	1.35e-55	7.287e-28	73.8%	0.073

Statistical Power vs. Robustness: Understanding Method Discordance

The striking discordance between GAM and SRH results (75.0% significant in LM only, 0.0% in rank test only, 2.6% both methods significant) reflects a fundamental trade-off in statistical methodology: **parametric methods maximize power when assumptions hold, while nonparametric methods sacrifice power for robustness to assumption violations.**

GAM's superior power derives from two factors:

1. Distributional assumptions: GAM assumes approximately normal residuals and homogeneous variance, which are reasonable after appropriate transformation for entropy data. Under these assumptions, parametric methods are theoretically optimal, achieving maximum power for a given type I error rate. The rank-based SRH test is assumption-free but necessarily discards quantitative information by converting measurements to ranks, which reduces statistical power when the underlying data are approximately normal.
2. Model flexibility with penalty: GAM uses thin-plate splines with smoothness penalties that simultaneously fit nonlinear patterns while controlling degrees of freedom. This flexibility allows GAM to detect subtle entropic index $\times$ group interactions across all q-values. The SRH test, by contrast, operates on rank patterns only, which is inherently less sensitive to continuous relationships across the ordering (q-values).

This complementary approach—combining high-power parametric tests with robust nonparametric alternatives—provides confidence that discoveries are methodologically sound rather than artifacts of statistical assumptions.

Visualization: Method Comparison

Visual comparison of concordance patterns provides intuitive assessment of which genes align between methods and which are method-specific. The following plots display significance calls, effect sizes, and agreement metrics in a format that facilitates interpretation of both robust discoveries and method-specific findings.

```

# Create comparison plot using S4 wrapper
# Concordance analysis results are stored in analysis metadata after calculate_concordance()
p <- plot_concordance(analysis, verbose = TRUE)
print(p)

```

Comparing interaction detection across two statistical methods

Concordance analysis between GAM and Scheirer-Ray-Hare tests

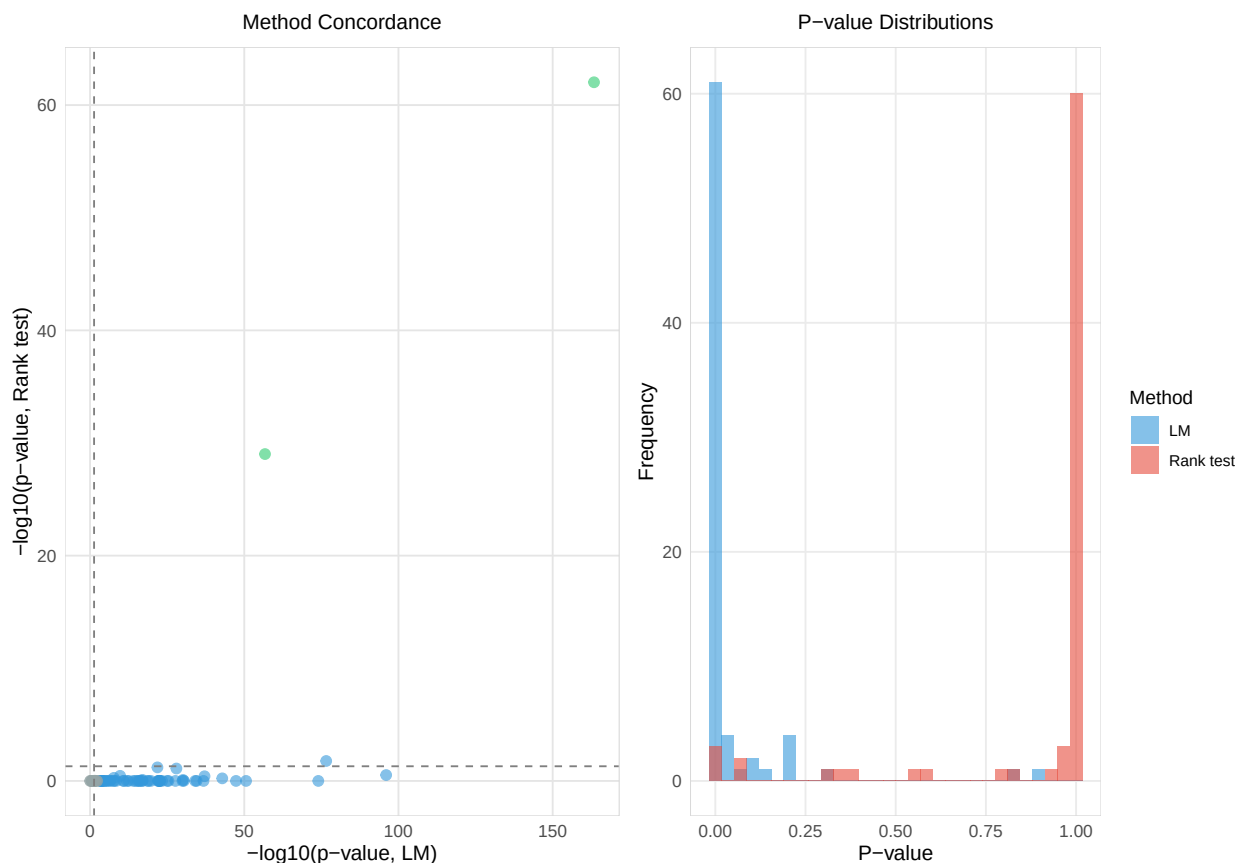


Figure 2: Method concordance visualization comparing SRH rank-based and GAM approaches for detecting $q \times \text{condition}$ interactions.

Session Information

```
sessionInfo()
#> R version 4.5.2 (2025-10-31)
#> Platform: x86_64-conda-linux-gnu
#> Running under: Ubuntu 22.04.5 LTS
#>
#> Matrix products: default
#> BLAS/LAPACK: /home/nouser/miniconda3/lib/libopenblas-r0.3.30.so; LAPACK version 3.12.0
#>
#> locale:
#>  [1] LC_CTYPE=es_ES.UTF-8      LC_NUMERIC=C
#>  [3] LC_TIME=de_DE.UTF-8      LC_COLLATE=es_ES.UTF-8
#>  [5] LC_MONETARY=de_DE.UTF-8  LC_MESSAGES=es_ES.UTF-8
```

```

#> [7] LC_PAPER=de_DE.UTF-8      LC_NAME=C
#> [9] LC_ADDRESS=C              LC_TELEPHONE=C
#> [11] LC_MEASUREMENT=de_DE.UTF-8 LC_IDENTIFICATION=C
#>
#> time zone: Europe/Berlin
#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats4      stats      graphics  grDevices  utils      datasets  methods
#> [8] base
#>
#> other attached packages:
#> [1] gridExtra_2.3              ggplot2_4.0.2
#> [3] knitr_1.51                 dplyr_1.2.0
#> [5] SummarizedExperiment_1.40.0 Biobase_2.70.0
#> [7] GenomicRanges_1.62.1      Seqinfo_1.0.0
#> [9] IRanges_2.44.0            S4Vectors_0.48.0
#> [11] BiocGenerics_0.56.0       generics_0.1.4
#> [13] MatrixGenerics_1.22.0     matrixStats_1.5.0
#> [15] kableExtra_1.4.0          TSENAT_0.99.0
#> [17] testthat_3.3.2
#>
#> loaded via a namespace (and not attached):
#> [1] tidyselect_1.2.1          viridisLite_0.4.3        farver_2.1.2
#> [4] S7_0.2.1                  fastmap_1.2.0            digest_0.6.39
#> [7] lifecycle_1.0.5          magrittr_2.0.4           compiler_4.5.2
#> [10] rlang_1.1.7               sass_0.4.10              tools_4.5.2
#> [13] yaml_2.3.12               labeling_0.4.3           S4Arrays_1.10.1
#> [16] pkgbuild_1.4.8            DelayedArray_0.36.0      xml2_1.5.2
#> [19] RColorBrewer_1.1-3        pkgload_1.5.0            abind_1.4-8
#> [22] BiocParallel_1.44.0       withr_3.0.2              purrr_1.2.1
#> [25] geepack_1.3.13           desc_1.4.3               grid_4.5.2
#> [28] scales_1.4.0             MASS_7.3-65              tinytex_0.58
#> [31] cli_3.6.5                rmarkdown_2.30           otel_0.2.0
#> [34] rstudioapi_0.18.0        tzdb_0.5.0               cachem_1.1.0
#> [37] stringr_1.6.0            splines_4.5.2            parallel_4.5.2
#> [40] XVector_0.50.0           vctrs_0.7.2              Matrix_1.7-4
#> [43] jsonlite_2.0.0           hms_1.1.4                systemfonts_1.3.2
#> [46] tidyr_1.3.2              jquerylib_0.1.4          glue_1.8.0
#> [49] codetools_0.2-20         cowplot_1.2.0            stringi_1.8.7
#> [52] gtable_0.3.6             tibble_3.3.1            pillar_1.11.1
#> [55] htmltools_0.5.9         brio_1.1.5               R6_2.6.1
#> [58] textshaping_1.0.5        rprojroot_2.1.1          evaluate_1.0.5
#> [61] lattice_0.22-9           readr_2.2.0              backports_1.5.0
#> [64] pheatmap_1.0.13          memoise_2.0.1            broom_1.0.12
#> [67] bslib_0.10.0            Rcpp_1.1.1               svglite_2.2.2
#> [70] SparseArray_1.10.9       nlme_3.1-168             mgcv_1.9-4
#> [73] xfun_0.56                pkgconfig_2.0.3

```